TECHNICAL DOCUMENT / ARTIFACT ANALYSIS

# Common metadata

Comparison of observable fields and independently reproduced checks across three SPARTACO artifacts.

| FIELD | rol_PRIV.png | rol_KIT.png | SC_IMG_9115.jpeg.spartaco |
|---|---|---|---|
| Visible schema / version | 1 / 2 · PNG | 1 / 2 · PNG | 2 / 2 · envelope |
| Encrypted envelope | 2 / 2 | 2 / 2 | 2 / 2 |
| Sealed (UTC) | 2026-10-12 22:55:40 | 2026-10-12 22:55:40 | not exposed |
| Artifact ID | `ART-c12c575a…83e3eb` | `ART-7c273f74…7814a2` | `ART-2fa0d417…5c0ce4` |
| Authority ID | ed25519:0f63a34d… | ed25519:0f63a34d… | ed25519:0f63a34d… |
| Encryption | AES-256-GCM | AES-256-GCM | AES-256-GCM |
| KDF | HKDF-SHA256 | HKDF-SHA256 | HKDF-SHA256 |
| Declared suite (raw) | `SPARTACO/1 \| apartado: firma=Ed25519 \| hash/canales=SHA-256(S,Q por mitades) \| kdf=HKDF-SHA256(dominios AUTH/DEF/DOC/REC) \| cifrado=AES-256-GCM(nonce ligado al canal) \| eje=3` | | |
| Ciphertext · bytes | 2,497 B | 3,000 B | 225,880 B |
| ciphertext_digest | d8f088c180…10e63635 OK | 52e9540b96…b75d7b2f OK | 6eb657d7de…a10ed5ab OK |
| nucleo / S / Q | SHA-256 OK in signed blob | SHA-256 OK in signed blob | S/Q not exposed; plaintext_nucleo declared |
| tag_auth / tag_def | both present | both present | both present |
| wrap_recuperacion | present | present | present |
| Ed25519 signature (outer) | VALID with rol_PUB.pub | VALID with rol_PUB.pub | not exposed |
| VPSI seal / marker | `#VPSI-NEUTRO:5` | `#VPSI-NEUTRO:2` | `#VPSI-NEUTRO:0` |
| n_neutro · vector | `3 · [4,1,0,0,0,0,1,0]` | `3 · [0,2,4,0,1,1,0,0]` | `not exposed` |

### VERIFIED WITHOUT SECRETS

PRIV and KIT Ed25519 signatures, public-key fingerprint, and blob/S/Q hashes verified. Ciphertext digests match in all three files.

### SCOPE OF THE EVIDENCE

Authentication tags and recovery wrapping are observed as fields only. No content was decrypted, and exclusive access was not established.

RESULTS / INTEGRITY AND AUTHENTICITY

# Verification and evidence

Direct cryptographic checks performed using the public key and the original artifact bytes.

**2 / 2**
Ed25519 signatures
valid canonical bodies

**2 / 2**
nucleo · S · Q
blob hashes matched

**3 / 3**
ciphertext SHA-256
digests recomputed

PUBLIC KEY · SHA-256 (32 BYTES)

rol_PUB.pub

`ed25519:0f63a34dce0fba561495b5ce16d5373875d666cf55e77c45dd494fb7eb291093`

## IDENTITY PACKAGES / OUTER MANIFESTS

### rol_PRIV.png

ED25519 · VALID SIGNATURE

artifact_id
`ART-c12c575ae9ec43ac3a2f7120d083e3eb`

blob: 3,497 bytes · ciphertext: 2,497 bytes

nucleo (SHA-256)
`5e356d12f05adc0188e3a02b2ef824ed`
`03454e61e36954c325d2c0847d6c8d01`

S (first half)
`16a25785aa304e65c0bb63df838bbbb0`
`d626a94508f44b5c8257519fae1a2305`

Q (second half)
`1988ec0de394284e27766e286ca10595`
`63e759529a86ccd22ee6fd8f45018654`

Signed manifest + hash-linked blob

### rol_KIT.png

ED25519 · VALID SIGNATURE

artifact_id
`ART-7c273f74089be82574dcf474867814a2`

blob: 4,000 bytes · ciphertext: 3,000 bytes

nucleo (SHA-256)
`17904dc7d915957956d4c18e3067f750`
`a9a8e6e9d5dafe87edb359b35f261822`

S (first half)
`1c96c46dd8bfbfb7890431ee43a30998`
`122896d11b726dd10a4ab6611d559a82`

Q (second half)
`fddb2bc5ac331f32c8f1173b8a571c6b`
`0f7802ce1dbd0b91d9de23f38f56b6fa`

Signed manifest + hash-linked blob

## DOCUMENT ENVELOPE / SC_IMG_9115.jpeg.spartaco

artifact_id `ART-2fa0d4177d34a5076a5c0ce4`

ciphertext_digest SHA-256
`6eb657d7de5847b8f3bb339be9537c657aed2016b69962d2ca3d41e4a10ed5ab`

CIPHERTEXT: 225,880 B · MATCHING HASH · NO OUTER SIGNATURE

## INTERPRETATION CRITERIA

Each signature validates the canonical body of its outer manifest. The nucleo / S / Q digests bind each PRIV/KIT blob to that body; this does not mean the PNG pixels are signed. All three ciphertext digests were recomputed over ciphertext bytes, excluding the separator and trailing marker.

Not evaluated: AES-256-GCM authentication with session keys, K_DOC unlocking, runtime validation of tag_auth / tag_def, or end-to-end execution of PROTECCION.verificar. This metadata does not establish whether the content can be opened.